

RFM Segmentation Summary

1. Pipeline Overview (What was done)

- Data scope: Built customer segmentation from train_processed.parquet only to enforce strict no-leakage.
- Feature engineering (Polars): Aggregated customer-level RFM from active training transactions.
- Definitions: $R = \min(\text{days_ago})$, $F = \text{count}(\text{events})$, $M = \text{sum}(\text{price} * \text{quantity})$.
- Preprocessing: Applied Yeo-Johnson to R/F/M, then StandardScaler to transformed R/F/M plus age.
- Clustering: Trained KMeans (K=5) after K-search diagnostics.
- Business mapping: Converted clusters to interpretable segments using rank-based deterministic labeling (Whales, Champions, At-Risk, Newcomers, Core + Young/Mature age prefix).

2. Reliability & Quality Metrics (Why trust this?)

- Coverage proof: 1,338,785 output rows, exactly matching 1,338,785 unique active train customers.
- Integrity proof: 0 nulls, 0 NaNs, 0 duplicate customer_id in final artifact.
- Method proof: Elbow + Silhouette were computed on a 100k sample; K=5 was chosen for better business-actionable granularity versus the raw silhouette peak at K=3.
- Memory optimization: Export schema uses customer_id Int32, cluster_id Int32, age Int8 to reduce footprint and keep downstream joins stable.

3. Segment Profiles & Results (The Output)

- Young At-Risk (Share: 24.25% | Medians: R=460, F=3, M=0.086373, Age=27 | Basket Size: 1.0994 items)

Description: Low-frequency, stale customers with very weak recent activity. Prioritize win-back triggers and high-CTR staples.

- Young Whales (Share: 22.12% | Medians: R=72, F=37, M=1.036119, Age=27 | Basket Size: 1.1079 items)

Description: Highest-value and most active buyers. Prioritize premium cross-sell and margin-maximizing bundles.

- Mature Newcomers (Share: 20.06% | Medians: R=385, F=3, M=0.101627, Age=53 | Basket Size: 1.0941 items)

Description: Low-history older customers with limited engagement depth. Use onboarding and age-relevant trend recommendations.

- Young Core (Share: 18.05% | Medians: R=94, F=7, M=0.186390, Age=25 | Basket Size: 1.0770 items)

Description: Broad mid-value base with moderate repeat behavior. Use frequency-lift offers and personalized replenishment.

- Mature Champions (Share: 15.52% | Medians: R=84, F=26, M=0.779186, Age=51 | Basket Size: 1.1033 items)

Description: Loyal, strong spenders with consistent activity. Target curated assortments and premium retention campaigns.

4. Downstream Usage (For RecSys / MBA)

- RecSys: Treat segments as policy layers: prioritize high-margin candidate generation for Whales/Champions, and trend-plus-demographic cold-start ranking for Newcomers.
- MBA: Use segment-level avg basket size to tune FP-Growth support thresholds (lower for sparse groups like Core, stricter confidence for premium groups).